

Thesis Defense Transcript

Data-Driven Cognitive Phenotyping in Acquired Brain Injury

20-minute presentation script

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Target duration: 20 minutes

Slide 1. Title and Introduction

Time on this slide: 45 seconds.

Hi. Welcome. The title of my dissertation is “Data Driven Cognitive Phenotyping in Acquired Brain Injury.”

The principal issue addressed in my dissertation is whether routinely recorded neuropsychological data can serve as a consistent and reliable cognitive classification system after acquired brain injury (ABI), even though the neuropsychological datasets typically are less well-defined than clinical datasets.

In short, yes. But the cognitive substructures identified by clustering did not represent clearly defined boundaries separating different cognitive subgroups. Rather, the most prominent signal detected was a cognitive severity gradient. Along a general dimension of cognitive performance, patients were located along this dimension. The clustering methodology subsequently separated this general dimension of cognitive performance into three distinct levels of clinically significant severity.

Note: Provide enough background about the overall argument before you begin discussing Methods.

Slide 2. Thesis in One Sentence

Time on this slide: 45 seconds.

My thesis in one sentence is: A clear cognitive severity gradient can be identified using unsupervised learning instead of a hidden subgroup structure in incomplete routine neuropsychological data sets.

There are three key ideas contained in this single sentence. First, diagnosis does not explain the variability in cognition fully. Two patients receiving the same high-level diagnosis for ABI may have differing needs for rehabilitation. Second, missingness must be taken into account through imputation, as many missing values in the data are not random missing fields in a table. Typically, they are indicative of fatigue, cognitive impairment, test intolerance, examiner judgment, or simply the physical constraints of assessing the degree of cognitive impairment. Third, no matter which strategy for imputation is chosen, the tiers that are identified are highly correlated. Therefore, it increases confidence that the results obtained from my clustering solution(s) are more credible than those resulting from an arbitrary preprocessing approach selected without regard to its influence on clustering. Therefore, my thesis addresses both a clinical and a methodological question. Clinically, I asked what type of cognitive organization occurs post-ABI. Methodologically, I asked if that organization persists notwithstanding the necessity to make decisions regarding missing data to enable analysis of routinely collected clinical data.

Slide 3. Clinical Problem

Time on this slide: 1 min.

The clinical problem is characterized by heterogeneity. ABI can affect a wide array of cognitive functions including attention, memory, language, executive function, visuospatial perception and/orientation to name just a few. Furthermore, the cognitive impairments exhibited by patients do not occur in identical manners.

Clinicians rely heavily on broad categories such as diagnosis or severity of injury to describe patients' cognitive challenges. These categories provide a valuable conceptual framework for appreciating the range of potential rehabilitation needs for patients. Nonetheless, these categories do not possess sufficient detail to allow clinicians to differentiate rehabilitation needs based on an individual patient's specific pattern and degree of cognitive difficulty. For example, two patients exhibiting memory deficits may exhibit substantially disparate patterns and degrees of cognitive difficulty.

Secondarily, the clinical problem includes missingness. Frequently, neuropsychological test batteries include substantial gaps as a result of the reality that neuropsychologists often assess patients during times when they are least able to successfully complete complex battery tests. When all incomplete cases are excluded from analysis, essentially half (approximately 49.9%) of all assessments made by patients whose treatment plans would likely benefit most from a comprehensive description of their cognitive abilities would be removed from consideration in the analysis.

Therefore, my research question is: Can data-driven clustering generate clinically meaningful cognitive organizational schemes? Moreover, if data-driven clustering generates clinically meaningful organizational schemes, can I trust that such organizational schemes will persist even when missing data are properly handled rather than being ignored?

Slide 4. Pre-registered Tests

Time on this slide: 1 minute.

I developed my analysis plan centered around four quantitative hypotheses. Hypothesis one (H1) tested whether discrete and stable clusters existed. To establish whether that was true, I outlined several pre-specified criteria to evaluate that question. Those criteria consisted of an average silhouette value > 0.40 and $< 30\%$ noise for virtually all of the imputation strategies examined. Additionally, evidence supporting the existence of at least two clusters was necessary for the majority of the imputation methods examined.

Hypothesis two (H2) evaluated the stability of clustering to imputation. Similar to H1, I specified a threshold for establishing stability. That is to say that I desired an average pairwise adjusted Rand Index $> .50$ across imputation methods. Note once again that H2 utilized a fixed reference design. As such, it employed a single fixed reference for evaluating similarity - i.e., the embedding and partition remained unchanged while the values to be imputed were permitted to vary.

Hypothesis three (H3) tested whether the tiers identified through clustering possessed clinical significance. If the tiers are clinically significant, they ought to be related to clinical unit classifications. I conducted this inquiry using chi-square association statistics, Cramer's V coefficient and adjustment for diagnosis group.

Lastly, hypothesis four (H4) tested whether aggregating variables at the domain level resulted in better separation than utilizing all individual variables. I compared separation metrics, variance inflation factor coefficients and condition numbers to accomplish this goal.

Cue: Be confident when providing this explanation to the examiners. Avoid overly detailing statistical formulas here - provide them simply so that the examiners understand that there is an objective test of your assertions provided throughout this presentation.

Slide 5. Cohort

Time on this slide: 1 minute.

Although my cohort is relatively large, it is incomplete; that is why this dissertation required both a clustering methodology and a missing-data strategy.

All analyses were performed using data from a rehabilitation center neuropsychology database. Following tier-1 filtering and eliminating assessments that lacked any type of cognitive testing from consideration, there were 17,406 assessments from 7,285 patients available for examination. Of those variables assessed in relation to six cognitive domains, only 14 variables were retained for subsequent analyses.

Only 53.1 percent of rows in the dataset had complete values across all 14 variables. Complete-case analysis would discard nearly half (approximately 46.9%) of all assessments in a clinical rehabilitation database. Discarding almost half of the assessments in a clinical rehabilitation database would significantly impact the characteristics of the patient population analyzed.

Therefore, I view missingness as non-randomly distributed across the variables assessed; hence, missingness cannot be viewed as an ancillary characteristic of the datasets analyzed. Missingness reflects various aspects of the clinical testing process. Assessments completed earlier in testing may reflect fatigue or decreased motivation; assessments completed later may reflect cognitive overload or task aversion; assessments omitted may reflect either insufficient test tolerance or poor decision-making on behalf of clinicians relative to choosing alternative forms of assessment; etc.

However, if all incomplete rows are discarded to create a cleaner numeric representation of the dataset; then that action reduces the probability that patients with severe cognitive impairment will be included in analyses examining associations between cognition and outcomes. Therefore, analyses must utilize imputation strategies designed to maintain partially observed assessments while allowing for assessment of uncertainty/sensitivity regarding missing values.

Slide 6. Missingness

Time on this slide: 1 minute.

This slide offers additional rationale for addressing missingness as structured as opposed to random.

Screening tests administered routinely are much more complete than specialized assessments requiring extended amounts of time to administer.

As mentioned above, missingness reflects aspects of the clinical testing process and not simply random error in administration. Examples of contributing factors to missingness include fatigue, disorientation/confusion caused by motor deficits or cognitive load or decreased motivation/task aversion or clinician judgments that another test may be more appropriate for evaluating certain aspects of a patient's cognition.

Thus, missingness can affect the ability to define patients' rehabilitation requirements in terms extending beyond simply diminishing sample size. Removing partially observed rows will also tend to remove patients with higher degrees of cognitive deficit.

Therefore, analyses must employ imputation methodologies enabling retention of partially observed assessments while permitting evaluations of uncertainty/sensitivity relating to missing values.

Slide 7. Pipeline

Time on this slide: 1 minute and 20 seconds.

Below is illustrated my overall pipeline flow chart.

Illustrated in the flow chart is where I employed imputation methodologies to manage missingness in my datasets. Ten different imputation strategies ranging from simple/basic to model-based/donor-based/matrix completion/machine learning/deep learning approaches were evaluated.

After employing imputation methodologies, I normalized/stabilized scores across the imputed datasets and categorized/grouped them by six cognitive domains. The imputed datasets were subsequently embedded using Uniform Manifold Approximation and Projection (UMAP). Finally, hierarchical density-based spatial clustering of applications with noise (HDBSCAN) was used to determine clusters in embedded space. Separation vs percentage of noise in each cluster configuration evaluated was determined by calculating silhouette scores as a quality measure.

For hypothesis two (H2), I employed a fixed reference design. Utilizing Multiple Imputation By Chained Equations (MICE), I designated an anchor scaler/embedding/HDBSCAN cluster configuration (fixed reference) to project other imputation methods through this fixed structure. Importantly for H2, I tested whether imputed values reside in similar locations given a fixed structure versus separately fitting a full UMAP + HDBSCAN pipeline produces equivalent hierarchies

Slide 8. H1 results

Estimated duration: 1 min.

three cognitive severity tiers are evident. Labels for tiers were defined as above average (aa); near normal (nn), and Global Impairment (gi). Aa contained 6,725 assessments. Patients scoring higher than the cohort mean across domains. Nn contained 6,328 assessments. Situated near the cohort average. Gi contained 4,353 assessments. Below the cohort mean across domains.

All 10 Imputation methods exceeded the pre-registered thresholds. For the mice reference solution, the full coverage silhouette was about .53. The recovered number of tiers was three. I am unable to understand these as natural kinds or sharp distinctions between phenotypes. Instead, they appear as ordered bands along a severity continuum.

Slide 9. Severity profile

Approximately 1 min.

first major finding: a significant cognitive severity gradient exists following acquired brain injury (abi)

Principal components analysis (pca): confirms trend in data. Pca explains nearly 56 percent of variance in data and all domains have positive loads. Loading pattern consistent with broad cognitive severity dimension

Most prominent signal in data not symptom specific dissociation or catalog of distinct subtypes. Most dominant signal is overall cognitive severity

Slide 10. H2 results

Time: approximately 1 minutes and 15 seconds.

Second major finding relates to robustness of imputed values

Each pair of methods from each comparison met predetermined criteria for agreement established prior to conducting analyses.

Mean adjusted Rand index (ARI) among pairs of methods was .71, based upon series of bootstraps with a 95% confidence interval (.70,.72). Thus indicated substantial agreement among method pairs. Of the 45 comparisons made to assess agreement, 44 survived holm's correction. Only one pair did

not meet predetermined threshold. That pair was KNN vs. SoftImpute, which had a pairwise ARI just short of predetermined threshold.

High-confidence consensus labels were assigned to more than 86% of all assessments. This suggests little variation between selected Imputation methods relative to fixed embedding and partition. However, scope of claim only extends to demonstrate point assignments are relatively robust under fixed embedding and partition not independent clustering pipelines fitted to different datasets will produce identical density hierarchies.

Slide 11. Why Imputation matters

Time: approximately 1 minute.

Illustrated necessity for Imputation analysis. Complete cases represent only 53.1 percent of original 17,406 assessments.

When evaluating similarity between mice reference solution and baseline complete case solution, both solutions produced two clusters rather than desired three-tiered solution. Silhouette values also indicate baseline solution produces poorer silhouette values (about .26) compared to those produced by mice reference solution (about .53). Lastly, although ARI comparing the two solutions on common assessments was about .63, still represents less than perfect agreement.

Thus deleting cases due to missing data does not merely produce less-powerful version of same analysis. It changes partition. How missing data are handled forms part of model's output regarding its substantive outcome.

Slide 12. H3 result

Time: approximately 1 minute.

third hypothesis examines whether cognitive severity levels identified in previous slides relate to clinical unit assignment. Chi-square test between tier membership and clinical unit assignment yields a chi-square statistic equal to 911 and estimated p value greater than 10^{-156} . Cramer's $v = .16$ indicates small-to-moderate effect size according to standard interpretations. As per pre-registration document, designated an effect size of .10 as a threshold. Since cramer's $v = .16 > .10$ consider this result supporting h3.

Also employed cochrane-mantel-haenszel test to determine if clinical unit assignment can be solely explained by diagnoses. Even after controlling for diagnosis in cmh test, relationship remained statistically significant. Conclude while diagnostic classification plays role in determining clinical unit assignment, cognitive severity tier provides additional predictive power above-and-beyond diagnostic classification.

From a clinical perspective reasonable to assume individuals classified as having Global Impairment will require more intensive structured rehabilitative interventions than their peers who are not classified in this manner. Do not mean tier membership predetermines placement decisions in rehabilitation settings. Propose tier membership contributes valuable information to inform rehabilitation routing decisions.

Slide 13. H4 result

Time: approximately 1 minute.

Final hypothesis assesses if cluster solutions derived from aggregating domain-level scores perform comparable to cluster solutions generated using unaggregated (ie individual test variable) scores.

Domain level clustering performs marginally better than unaggregated individual test variable clustering. Silhouette values indicate marginal improvement in clustering quality (.48 for domains vs. .47 for Individual Test Variables). Additionally, when evaluating numerical stability via variance inflation factors (vifs), mean vif decreases significantly (from ~2.80 to ~1.90) and condition numbers improve dramatically (from ~62 to ~12).

Improvements are especially relevant for downstream applications involving distance-based clustering algorithms (eg UMAP, hdbscan), given such algorithms often highly sensitized to redundancy/correlation between features. By aggregating individual test scores into clinically meaningful domains, reduced feature redundancy and retained interpretable cognitive constructs.

As such, domain level clustering offers advantages in terms of computational stability and interpretability making it preferable to unaggregated clustering approaches when attempting to identify cognitively-distinctive patient populations.

Slide 14. Result Summary

Duration of this slide: approximately 45 seconds.

Summary of four tested hypotheses

Hypothesis #1 finds support: recovered stable clusters using unsupervised techniques but understand these as representing distinct levels of cognitive severity rather than categorically-distinct “subtypes.”

Hypothesis #2 finds support within context of fixed-reference design: average pairwise ARI ~ .71 suggests robust agreement among method pairs.

Hypothesis #3 finds support: severity tier related to clinical unit assignment and continued statistical significance even after controlling for diagnosis.

Hypothesis #4 finds support: domain level clustering exhibits superior numerical properties (silhouette & vif) compared to unaggregated test variable clustering.

Collectively findings provide empirical support for developing and applying missingness aware severity stratification approach for analyzing routine neuropsychological assessments collected post abi.

Slide 15. Interpretation

Duration: approximately 1 min., 15 sec.

Primary interpretation offered here is continuum of cognitive severity gradients existent in dataset; rather than collection of discrete subtypes.

Understanding this distinction is important. An interpretation of subtypes suggests qualitative form(s) of cognitive profile(s) exist. On the otherhand, interpretation of phenomenon as example of cognitive severity gradient suggests differences in cognitive function lie almost exclusively along an overall continuum of cognitive severity across multiple domains.

Evidence provided throughout manuscript supports latter viewpoint. Evidence included radial plots which are indicative of concentricity among all groups examined via radar plots; large proportion (~50%) of total variance explained by pc1; all domains exhibit positive loads on pc1. Further evidence supporting view comes from examining conceptual nature of each tier examined during current investigation which can be interpreted analogously to bands extracted from larger continuum.

While it may be possible that domain-specific dissociations exist following brain injury (eg selective deficits in memory following tbi), evidence presented in current manuscript suggests that pipeline utilizing Imputation-robust domain-level severity stratified framework yields stronger signals regarding overall cognitive severity across multiple domains than any domain-specific dissociation.

Finally, evidence presented here highlights potential pitfalls in identifying patterns resembling phenotypic differentiation spurred by poor scaling (and/or inappropriate scaling), use of non-cognitive measures, omission of missing data.

Slide 16. Utility - Clinical and Analytical

Approximately 1 Minute

Two potential groups of people could utilize this methodology and subsequent applications:

Clinical Users. The severity tiers could also be utilized as another source of support for clinicians when making the determination of rehabilitation intensity, and possibly help clinicians report cognitive assessment results along with associated confidence levels that can help alleviate misinterpretation of borderline cases.

Analytical Users. The present manuscript illustrates an example missing-data-aware pipeline that can be reproduced and modified by researchers looking to examine relationships between clinical outcomes and cognitive severity following ABI.

Ultimately, the end goal of this dissertation will be to increase transparency about how missing data are treated within statistical analyses. As opposed to trying to ignore missing data or eliminate all incomplete observations from analyses, researchers will now be able to make missing-data decisions transparent and quantifiable in addition to evaluating the impact they have on the resulting analyses.

Slide 17. Application: CogDash

Approximately 45 seconds

In order to improve usability, I developed CogDash - a Streamlit web-app that utilizes portions of my analyses to create an interactive decision-making tool for clinicians.

The CogDash app contains tools such as cluster explorers for each patient; patient look-up; classification of new patients into clusters; summaries of tested hypotheses; and access to advanced analytical tools. Overall, CogDash was designed to promote collaboration/understanding between clinicians wanting to assess the cognitive function of their own patients, and analysts providing analytical services; it is NOT meant to automatically determine clinical decisions.

One possible reason why CogDash has value is because models applied in rehabilitation environments need to be auditable. Clinicians need to be able to view the data used and at what points confidence is low.

Slide 18. Limitations

Approximately 1 Min., 15 Sec.

There are many limitations to this study:

Firstly, data was collected from one location, so replication studies are required before we consider this structural representation useful across locations collecting post-ABI neuropsychological data.

Secondly, this analysis did not represent a complete longitudinal recovery model. Although repeated assessments provided some information on recovery trajectories, including time since injury added additional context; and direct modeling of recovery would require additional longitudinal designs.

Thirdly, the Imputation strategy rested heavily on assumptions related to MAR missing-data-generating processes; however, it cannot be ruled out that MNAR processes exist based solely on

observable data. Specifically, extreme MNAR analyses indicated that global impairment tier is robust while the boundaries around that tier are likely subject to MNAR processes.

Finally, there were no available functional outcomes (e.g., return-to-work; LOS) to assess for validation purposes; therefore, these types of functional outcomes would provide the most valid external validators for future research examining whether tier membership provides utility for guiding rehabilitation plans.

Slide 19. Conclusion

Approximately 1 Minute

I will summarize three findings:

Firstly, routinely-collected post-ABI clinical neuropsychological data collect significant amounts of cognitive-severity gradient.

Secondly, the method of imputing missing data represents substantial modeling decisions and affect the resultant structure(s), therefore ignoring missing data as part of a minor preprocessing procedure could lead to different results depending on characteristics of available data.

Thirdly, severity-tiered representations by domain level represent clinically-usable & numerically-stable representations representing benefits compared to un-aggregated clustering approaches when identifying cognitively-distinctive patient populations.

Therefore collectively, this dissertation supports neither strong evidence for distinct cognitive phenotypes existing in this sample population; nor does it negate their existence completely; however it does support developing & applying an Imputation-robust severity-stratification framework that can be both clinically-interpretable and analytically-evaluated.

Slide 20. Thank You!

Approximately 30 Seconds

Thank you for your time. I'd like to answer any questions you may have pertaining to either utilizing my codebase/Dashboards, clarifying technical details (Imputation design/clustering choices/etc.), or discussing potential implications for future research projects examining cognitive-function/rehabilitation planning post-ABI.

Optional Final Sentence if Room is Silent: Overall, my hope is that the analyses shown here identified something clinically-useful-but-more conservatively-defined as a cognitive-severity continuum after acquired-brain-injuries rather than distinct cognitive subtypes